

Churn Report Verify — Analytics Report

Business Analytics Report

Project: Churn Report Verify

Dataset: sample_customer_churn.csv

Analysis date: 2026-09-24

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Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **sample_customer_churn.csv** (300 rows, 12 columns) covering 2024-01-01 to 2024-12-28 to answer the business question: *Customer churn dataset to predict which customers will churn and understand retention drivers..*

Key verified findings

- The original dataset contained 0 missing value(s) and 0 duplicate row(s), which have since been cleaned.
- A Random Forest model was trained to predict **Churn** (classification). The model correctly classifies about 71.7% of cases (Accuracy).

Model reliability, in plain terms: The model correctly classifies about 71.7% of cases (Accuracy). F1 score (balancing precision and recall) is 0.713. ROC-AUC is 0.792 (1.0 = perfect separation, 0.5 = no better than chance).

Top recommendations

- The most common observed outcome for Churn is '1' (32 of 60 test cases).
- Among customers, the 'Tenure Months: Lower' group shows the highest observed rate (77.0%, n=100), compared to 'Tenure Months: Higher' (33.0%, n=100).

2. About the Dataset

`sample_customer_churn.csv` contains 300 rows and 12 columns. It spans the period 2024-01-01 to 2024-12-28 (based on Signup Date).

Business question: Customer churn dataset to predict which customers will churn and understand retention drivers.

Column dictionary

Column	Type	Missing	Notes
Customer Id	Identifier	0.0%	
Age	Numeric	0.0%	typical range 18.00–75.00
Region	Category	0.0%	most common: West
Plan	Category	0.0%	most common: Standard
Contract Type	Category	0.0%	most common: Month-to-month
Tenure Months	Numeric	0.0%	typical range 1.00–72.00
Monthly Charge	Numeric	0.0%	typical range 15.27–118.69
Total Spend	Numeric	0.0%	typical range 75.45–8,521.1
Support Tickets	Numeric	0.0%	typical range 0.00–10.00
Satisfaction Score	Numeric	0.0%	typical range 1.00–4.90
Signup Date	Date/Time	0.0%	
Churn	Numeric	0.0%	typical range 0.00–1.00

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
300	0	0	1

Outliers

Column	Finding	Affected
total_spend	'total_spend' has 8 statistical outlier(s) outside [-2921.19, 7084.38].	8

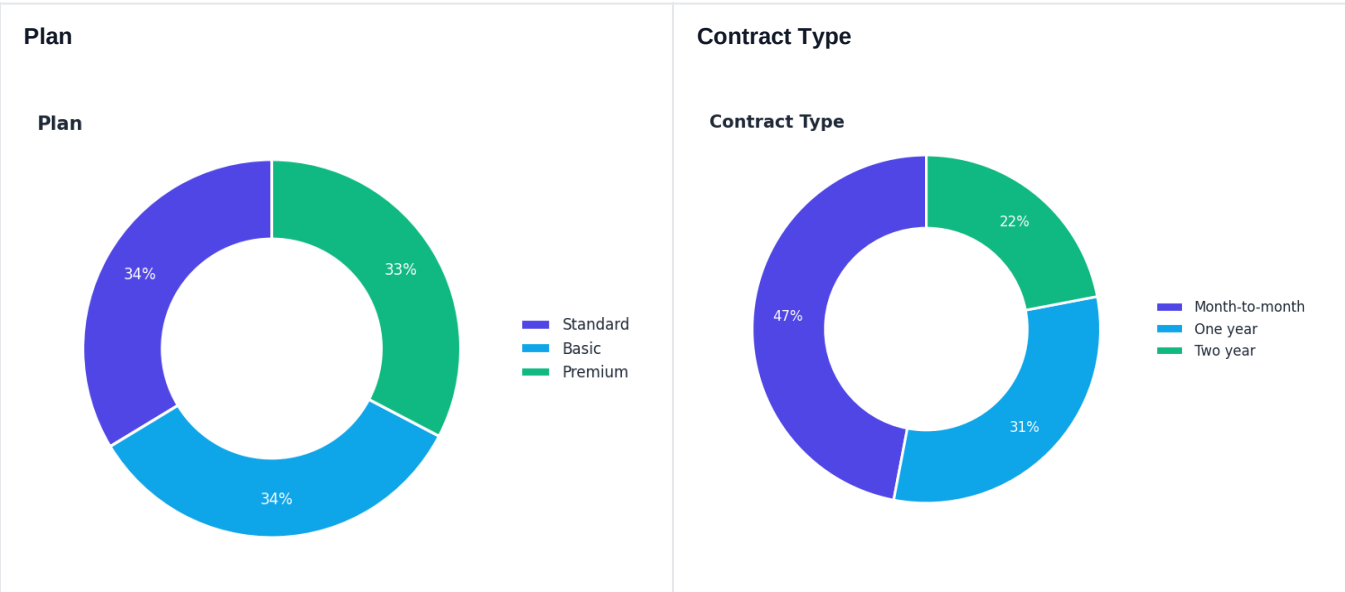
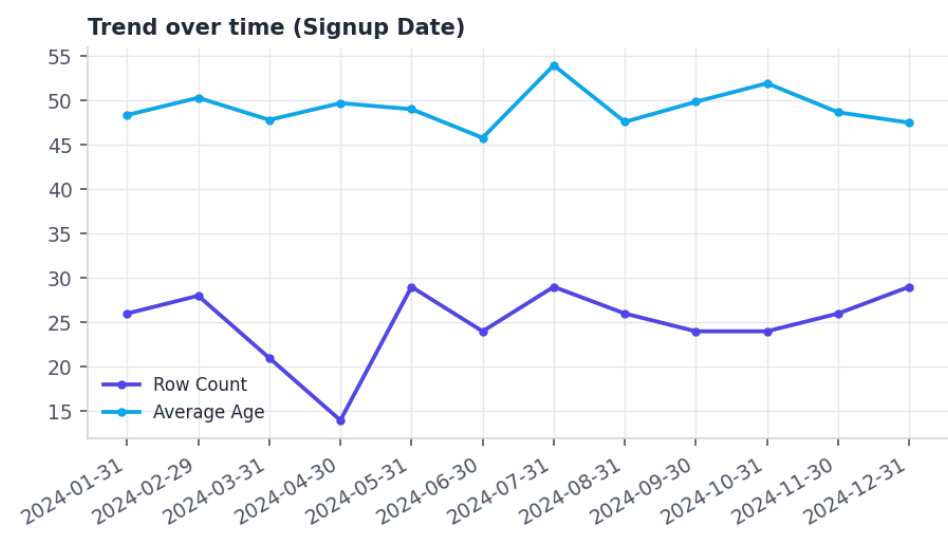
Cleaning performed

The dataset was marked cleaned, but no specific actions were logged.

4. Key EDA Findings

Total Rows	Churn Rate	Total Age	Total Tenure Months
300	53.7%	14,758	11,137

Trend over time (Signup Date)



Churn breakdown

Average: 0.54 (range 0.0–1.0)

Value	Count	Percent
1	161	53.7%
0	139	46.3%

5. Modeling Approach

A **classification** model was trained to predict **Churn**, using 11 feature(s).

Features used

Age, Region, Plan, Contract Type, Tenure Months, Monthly Charge, Total Spend, Support Tickets, Signup Date: Year, Signup Date: Month, Signup Date: Dayofweek

Features excluded, and why

Column	Reason excluded
Customer Id	Identifier column (e.g. an ID) — carries no predictive signal.
Signup Date	Raw date — replaced with derived year / month / day-of-week features instead.
Satisfaction Score	Excluded by default 'satisfaction_score' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (23% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below.

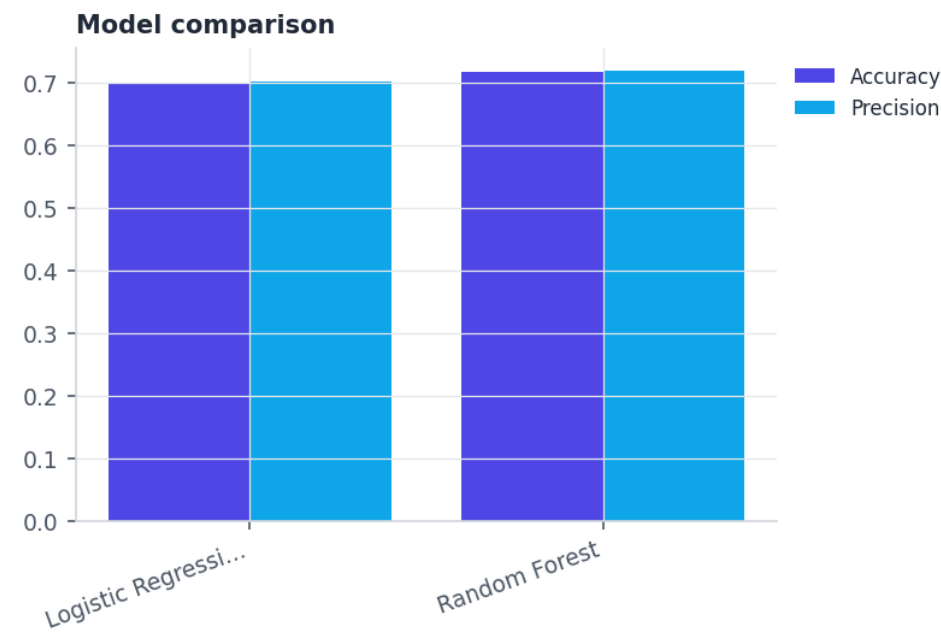
Technical details

Algorithm: Random Forest · Preprocessing: 8 numeric + 3 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.7000	0.7009	0.7000	0.6973	0.8136
Random Forest	0.7167	0.7197	0.7167	0.7130	0.7919

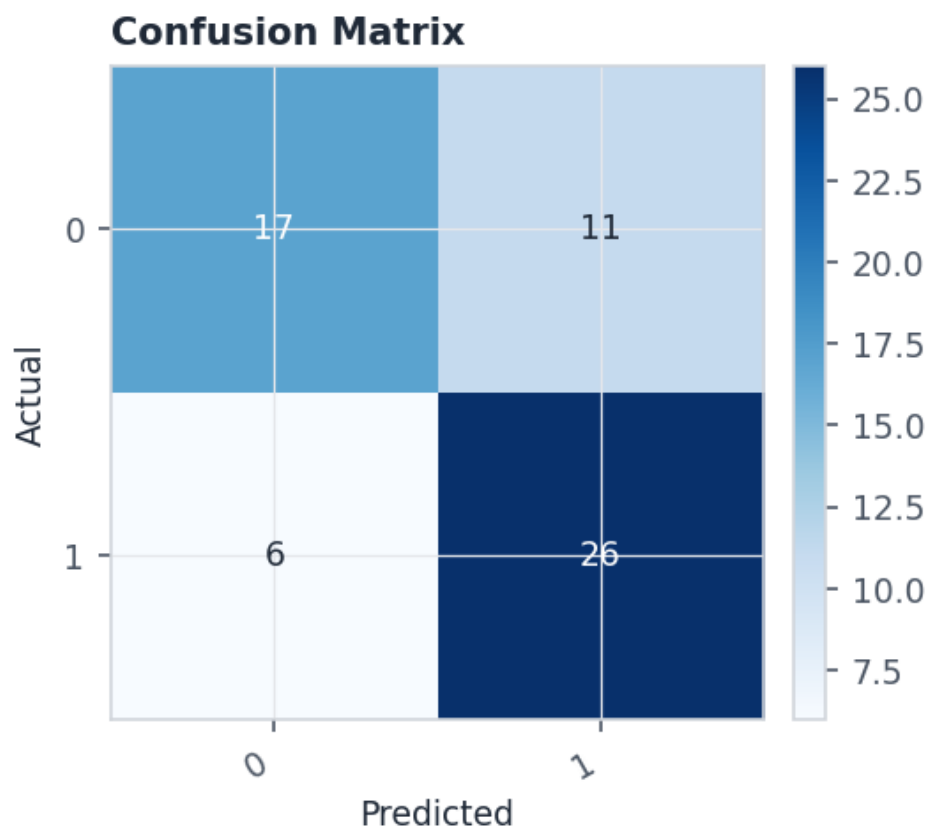


In plain language

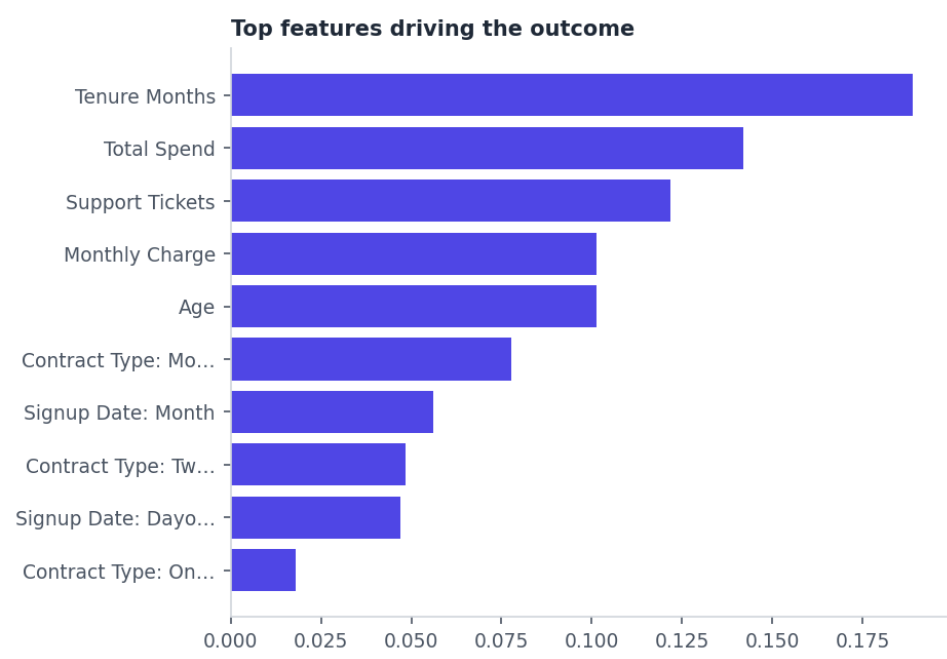
- The model correctly classifies about 71.7% of cases (Accuracy).
- F1 score (balancing precision and recall) is 0.713.
- ROC-AUC is 0.792 (1.0 = perfect separation, 0.5 = no better than chance).

Compared against a trivial baseline (F1 Score = 0.371): this model clearly outperforms it.

Confusion Matrix



7. What Drives the Outcome



Feature	Relative importance
Tenure Months	18.9%
Total Spend	14.2%
Support Tickets	12.2%
Monthly Charge	10.1%
Age	10.1%
Contract Type: Month to month	7.8%
Signup Date: Month	5.6%
Contract Type: Two year	4.8%
Signup Date: Dayofweek	4.7%
Contract Type: One year	1.8%

8. Future Outlook

- The most common observed outcome for Churn is '1' (32 of 60 test cases).
- Among customers, the 'Tenure Months: Lower' group shows the highest observed rate (77.0%, n=100), compared to 'Tenure Months: Higher' (33.0%, n=100).

Group	Observed rate	Group size
Tenure Months: Lower	77.0%	100
Tenure Months: Middle	51.0%	100
Tenure Months: Higher	33.0%	100

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- Cross-validation scores varied by ± 0.042 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Accuracy	0.7167
Precision	0.7197
Recall	0.7167
F1 Score	0.713
ROC-AUC	0.7919
Cv Mean	0.7648
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	2

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	300
Columns	12